Series 2 Survival Analysis

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(A)

Let the lifetimes (T_i) follow a Lognormal distribution:

$$\log T_i \sim N(\mu, \sigma^2).$$

Each observation consists of:

- (t_i): the observed time,
- (δ_i): event indicator
- $(\delta_i = 1)$: failure observed
- $(\delta_i = 0)$: right-censored observation

The Log-normal density is:

$$f(t; \mu, \sigma) = \frac{1}{t\sigma\sqrt{2\pi}} \exp\left(-\frac{(\log t - \mu)^2}{2\sigma^2}\right), \quad t > 0.$$

The survival function is:

$$S(t; \mu, \sigma) = 1 - \Phi\left(\frac{\log t - \mu}{\sigma}\right),$$

where Φ is the standard normal CDF.

The likelihood for right-censored data is:

$$L(\mu, \sigma) = \prod_{i=1}^{n} [f(t_i; \mu, \sigma)]^{\delta_i} [S(t_i; \mu, \sigma)]^{1-\delta_i}.$$

Log-Likelihood Function

$$\ell(\mu, \sigma) = \sum_{i=1}^{n} \{\delta_i \log f(t_i; \mu, \sigma) + (1 - \delta_i) \log S(t_i; \mu, \sigma)\}.$$

Expanding $\log f(t_i)$:

$$\log f(t_i; \mu, \sigma) = -\log t_i - \log \sigma - \frac{1}{2}\log(2\pi) - \frac{(\log t_i - \mu)^2}{2\sigma^2}.$$

Thus the full log-likelihood becomes:

$$\ell(\mu, \sigma) = \sum_{i=1}^{n} \delta_i \left[-\log t_i - \log \sigma - \frac{1}{2}\log(2\pi) - \frac{(\log t_i - \mu)^2}{2\sigma^2} \right] + \sum_{i=1}^{n} (1 - \delta_i) \log \left[1 - \Phi\left(\frac{\log t_i - \mu}{\sigma} \right) \right].$$

(B)

We have a table with observations for lifetime days of 20 components out of which the final 3 are right censored.

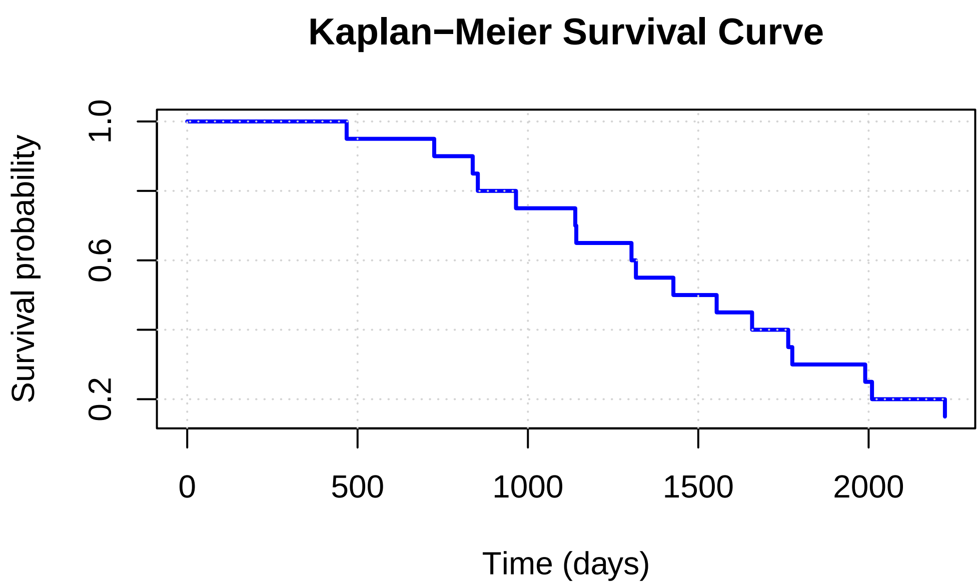
i) We are to calculate the Kaplan-Meier estimators and plot them:

```
time <- c(
  468, 725, 838, 853, 965,
  1139, 1142, 1304, 1317, 1427,
  1554, 1658, 1764, 1776, 1990,
  2010, 2224, 2244, 2279, 2286
)
status <- c(
  1, 1, 1, 1, 1,
  1, 1, 1, 1, 1,
  1, 1, 1, 1, 1,
  1, 1, 0, 0, 0
)

# Manual Kaplan-Meier calculation
n <- length(time)
ord <- order(time)
time_sorted <- time[ord]
status_sorted <- status[ord]

unique_times <- unique(time_sorted[status_sorted == 1])
km_est <- numeric(length(unique_times))
n_risk <- numeric(length(unique_times))
n_event <- numeric(length(unique_times))
```

```r
for (i in seq_along(unique_times)) {
  t <- unique_times[i]
  n_risk[i] <- sum(time_sorted >= t)
  n_event[i] <- sum(time_sorted == t & status_sorted == 1)

  if (i == 1) {
    km_est[i] <- 1 - n_event[i]/n_risk[i]
  } else {
    km_est[i] <- km_est[i-1] * (1 - n_event[i]/n_risk[i])
  }
}

# Create KM table
km_table <- data.frame(
  Time = unique_times,
  N_Risk = n_risk,
  N_Event = n_event,
  Survival = km_est
)

print(km_table)
```

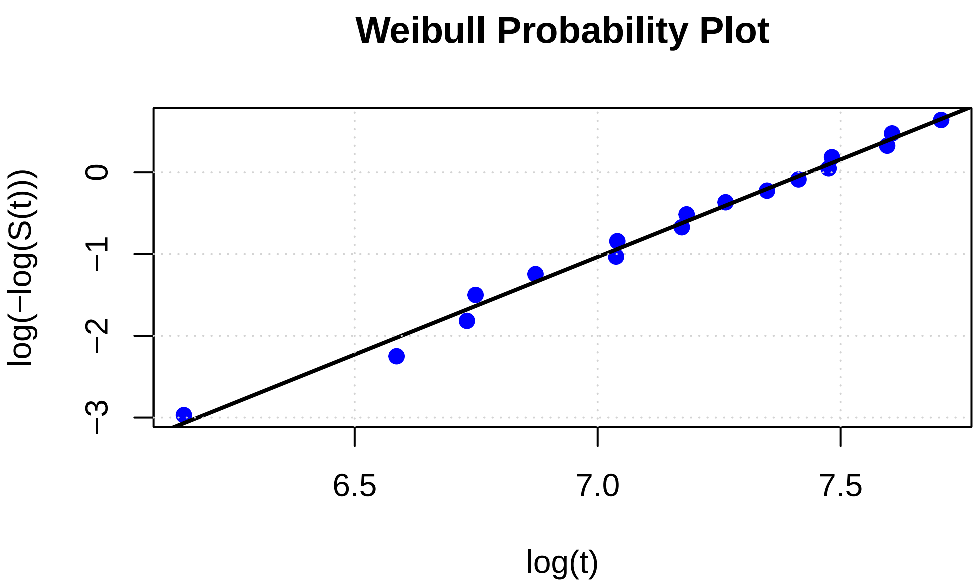
```
   Time N_Risk N_Event Survival
1   468     20       1     0.95
2   725     19       1     0.90
3   838     18       1     0.85
4   853     17       1     0.80
5   965     16       1     0.75
6  1139     15       1     0.70
7  1142     14       1     0.65
8  1304     13       1     0.60
9  1317     12       1     0.55
10 1427     11       1     0.50
11 1554     10       1     0.45
12 1658      9       1     0.40
13 1764      8       1     0.35
14 1776      7       1     0.30
15 1990      6       1     0.25
16 2010      5       1     0.20
17 2224      4       1     0.15
```

```
# Plot
plot(c(0, unique_times), c(1, km_est),
     type = "s",
     xlab = "Time (days)",
     ylab = "Survival probability",
     main = "Kaplan-Meier Survival Curve",
     lwd = 2, col = "blue")
grid()
```

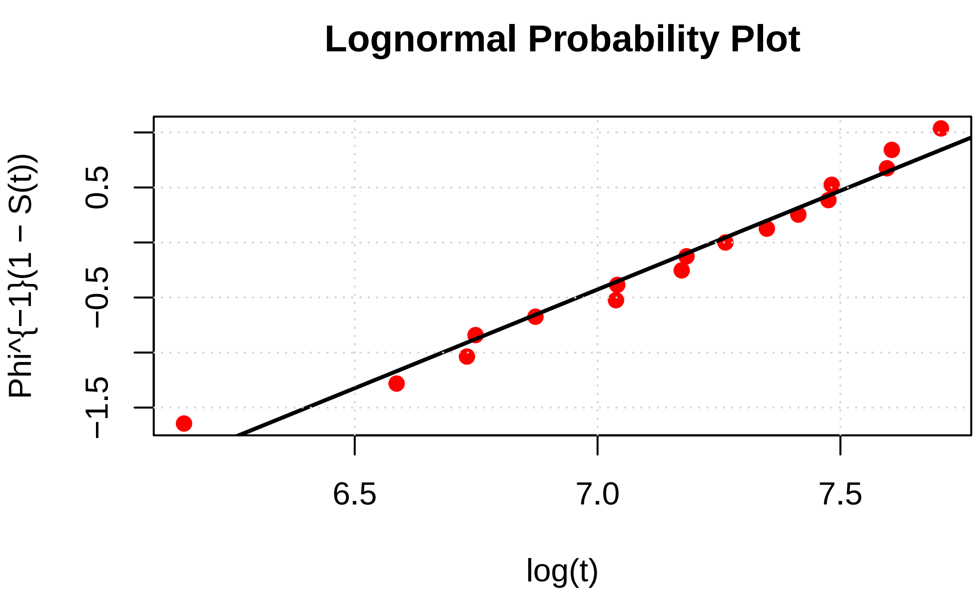


ii) We are now going to perform diagnostics to understand which parametric distribution (Weibull, Lognormal, Log-logistic) best fits the data.

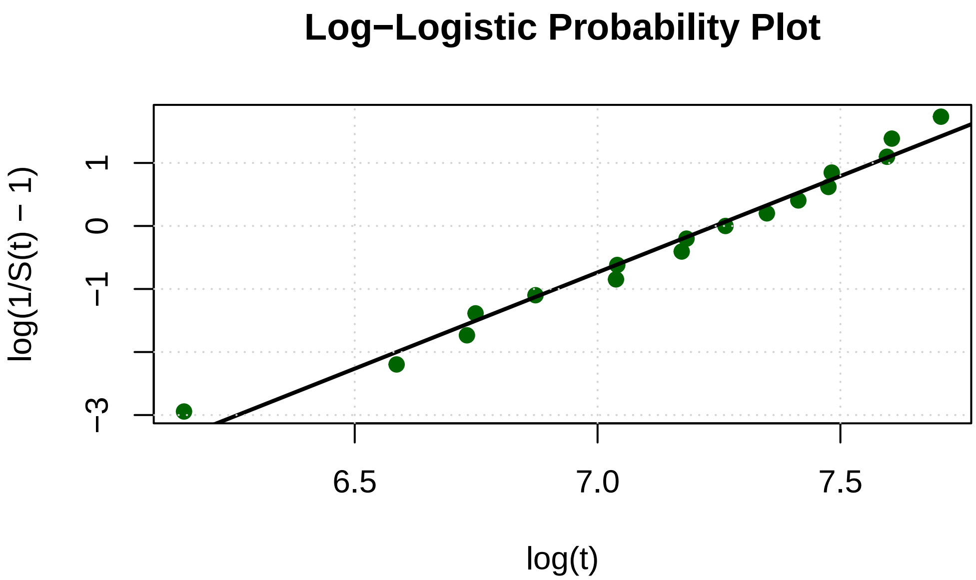
```
t <- km_table$Time
S <- km_table$Survival

X <- log(t)

### --- 1. Weibull Probability Plot ---
Y_weibull <- log(-log(S))

plot(X, Y_weibull, pch=19, col="blue",
     xlab="log(t)", ylab="log(-log(S(t)))",
     main="Weibull Probability Plot")
abline(lm(Y_weibull ~ X), col="black", lwd=2)  # fitted line
grid()
```




```r
### --- 2. Lognormal Probability Plot ---
Y_lognormal <- qnorm(1 - S)

plot(X, Y_lognormal, pch=19, col="red",
     xlab="log(t)", ylab="Phi^{-1}(1 - S(t))",
     main="Lognormal Probability Plot")
abline(lm(Y_lognormal ~ X), col="black", lwd=2)  # fitted line
grid()
```



```r
### --- 3. Log-Logistic Probability Plot ---
Y_loglogistic <- log(1/S - 1)

plot(X, Y_loglogistic, pch=19, col="darkgreen",
     xlab="log(t)", ylab="log(1/S(t) - 1)",
     main="Log-Logistic Probability Plot")
abline(lm(Y_loglogistic ~ X), col="black", lwd=2)  # fitted line
grid()
```



We are also going to show the R^2 values to compare:

```r
summary(lm(Y_weibull ~ X))$r.squared
```

```
[1] 0.9905424
```

```r
summary(lm(Y_lognormal ~ X))$r.squared
```

```
[1] 0.9714351
```

```r
summary(lm(Y_loglogistic ~ X))$r.squared
```

```
[1] 0.9791864
```

Based on the probability plots and the corresponding R^2 values from linear regressions, the Weibull model provides the best fit to the data. Its transformed KM points lie closest to a straight line, indicating that the Weibull distribution most accurately captures the underlying failure mechanism.

iii) We are now going to fit all these models to our data with the Max-Likelihood method.

```r
# Data
time <- c(
  468, 725, 838, 853, 965,
  1139, 1142, 1304, 1317, 1427,
  1554, 1658, 1764, 1776, 1990,
  2010, 2224, 2244, 2279, 2286
)

status <- c(
  1, 1, 1, 1, 1,
  1, 1, 1, 1, 1,
  1, 1, 1, 1, 1,
  1, 1, 0, 0, 0
)
```

Weibull Distribution

```r
negloglik_weibull <- function(par, time, status) {
  k      <- exp(par[1])
  lambda <- exp(par[2])

  t <- time
  d <- status

  # log f(t)
  logf <- d * ( log(k) - log(lambda) + (k - 1)*log(t/lambda) - (t/lambda)^k )

  # log S(t)
  logS <- (1 - d) * ( -(t/lambda)^k )

  out <- -(sum(logf) + sum(logS))

  if (!is.finite(out)) return(1e100)
  out
}
```

```
fit_weibull <- optim(
  par = c(log(1), log(median(time))),
  fn = negloglik_weibull,
  time = time,
  status = status,
  hessian = TRUE,
  method = "L-BFGS-B",
  lower = c(log(1e-6), log(1e-6)),
  upper = c(log(1e6),  log(1e6))
)

k_hat_weibull      <- exp(fit_weibull$par[1])
lambda_hat_weibull <- exp(fit_weibull$par[2])
```

Lognormal Distribution

```
negloglik_lognormal <- function(par, time, status) {
  mu  <- par[1]
  sig <- exp(par[2])

  t <- time
  d <- status

  z <- (log(t) - mu) / sig

  logf <- d * ( -log(t) - log(sig) - 0.5*log(2*pi) - 0.5*z^2 )

  S <- pmax(1 - pnorm(z), 1e-12)
  logS <- (1 - d) * log(S)

  out <- -(sum(logf) + sum(logS))

  if (!is.finite(out)) return(1e100)
  out
}


fit_lognormal <- optim(
  par = c(log(median(time)), log(1)),
  fn = negloglik_lognormal,
```

```
  time = time,
  status = status,
  hessian = TRUE,
  method = "L-BFGS-B",
  lower = c(-50, log(1e-6)),
  upper = c( 50, log(1e6))
)

mu_hat_lognormal  <- fit_lognormal$par[1]
sig_hat_lognormal <- exp(fit_lognormal$par[2])
```

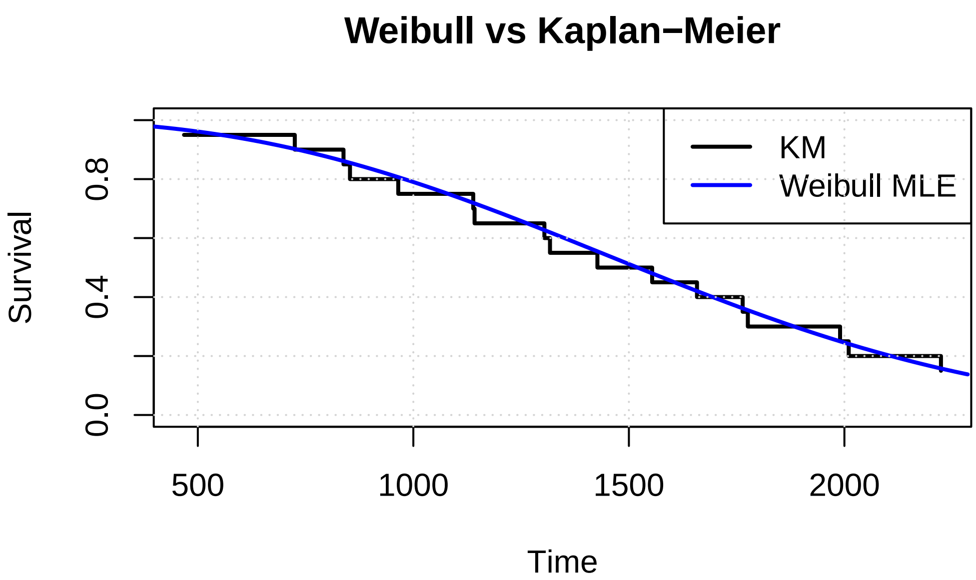
Log-Logistic Distribution

```
negloglik_loglogistic <- function(par, time, status) {
  alpha <- exp(par[1])
  beta  <- exp(par[2])

  t <- time
  d <- status

  z <- t / beta
  z_alpha <- exp(alpha * log(pmax(z, 1e-12)))

  logf <- d * ( log(alpha) - log(beta) + (alpha - 1)*log(pmax(z, 1e-12)) -
                2*log1p(z_alpha) )

  logS <- (1 - d) * ( -log1p(z_alpha) )

  out <- -(sum(logf) + sum(logS))

  if (!is.finite(out)) return(1e100)
  out
}


fit_loglogistic <- optim(
  par = c(log(1), log(median(time))),
  fn = negloglik_loglogistic,
  time = time,
  status = status,
  hessian = TRUE,
  method = "L-BFGS-B",
```

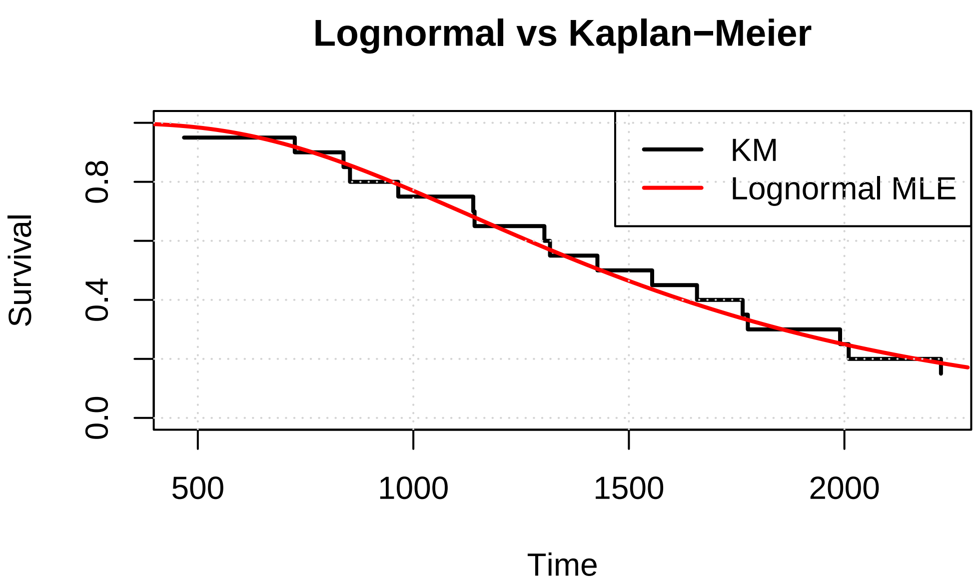
```r
  lower = c(log(1e-6), log(1e-6)),
  upper = c(log(1e6),  log(1e6))
)

alpha_hat_loglogistic <- exp(fit_loglogistic$par[1])
beta_hat_loglogistic  <- exp(fit_loglogistic$par[2])
```

We are going to do the plots:

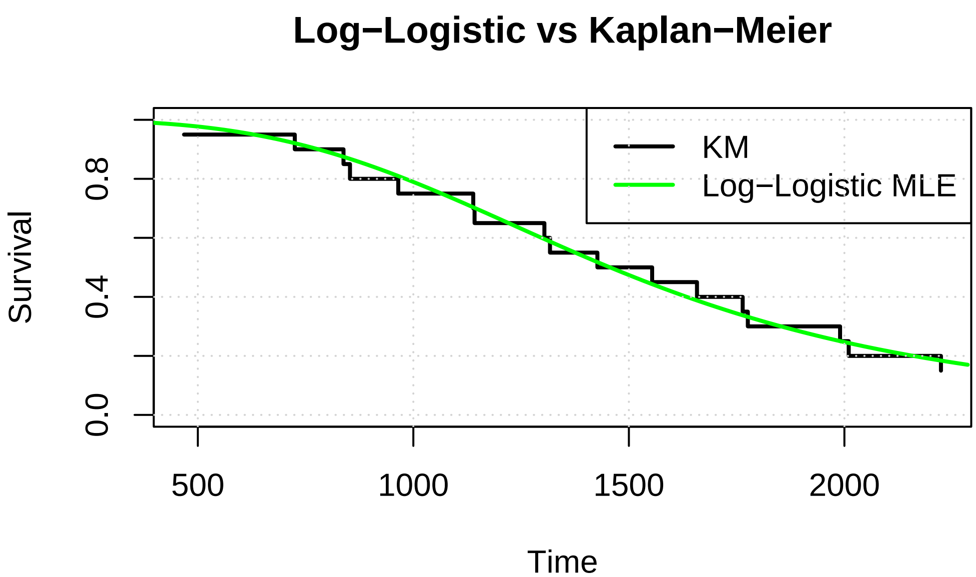
```r
t_seq <- seq(1e-6, max(time), length.out = 400)

plot_km <- function(km, ...) {
  plot(km$times, km$surv, type="s", lwd=2, col="black",
       xlab="Time", ylab="Survival", ...)
}



km_estimator <- function(time, status) {
  # Only failures contribute to jumps
  event_times <- sort(unique(time[status == 1]))

  n <- length(time)
  at_risk <- sapply(event_times, function(t) sum(time >= t))
  events  <- sapply(event_times, function(t) sum(time == t & status == 1))

  # KM survival
  surv <- cumprod(1 - events / at_risk)

  list(times = event_times, surv = surv)
}

km <- km_estimator(time, status)

S_weibull <- exp(-(t_seq/lambda_hat_weibull)^k_hat_weibull)

plot_km(km, main="Weibull vs Kaplan-Meier", ylim=c(0,1))

lines(t_seq, S_weibull, col="blue", lwd=2)

legend("topright", legend=c("KM", "Weibull MLE"),
       col=c("black","blue"), lwd=c(2,2))
grid()
```




```r
S_lognormal <- 1 - pnorm((log(t_seq) - mu_hat_lognormal)/sig_hat_lognormal)

plot_km(km, main="Lognormal vs Kaplan-Meier", ylim=c(0,1))

lines(t_seq, S_lognormal, col="red", lwd=2)

legend("topright", legend=c("KM", "Lognormal MLE"),
       col=c("black","red"), lwd=c(2,2))
grid()
```



```r
S_loglogistic <- 1 / (1 + (t_seq/beta_hat_loglogistic)^alpha_hat_loglogistic)

plot_km(km, main="Log-Logistic vs Kaplan-Meier", ylim=c(0,1))

lines(t_seq, S_loglogistic, col="green", lwd=2)

legend("topright", legend=c("KM", "Log-Logistic MLE"),
       col=c("black","green"), lwd=c(2,2))
grid()
```



We fitted the Weibull, Lognormal, and Log-logistic distributions to the lifetime data using Maximum Likelihood estimation. For each model, we computed the fitted survival curve $\hat{S}(t; \hat{\theta})$ and compared it visually with the non-parametric Kaplan–Meier estimator. The three plots (KM vs Weibull, KM vs Lognormal, KM vs Log-logistic) illustrate the agreement between the parametric models and the empirical survival function.

iv) We do the model comparison with the AIC:

```r
model_comparison <- data.frame(
  Model = c("Weibull", "Lognormal", "Log-Logistic"),
  LogLik = c(-fit_weibull$value,
             -fit_lognormal$value,
             -fit_loglogistic$value),
  AIC = c(2*2 + 2*fit_weibull$value,
          2*2 + 2*fit_lognormal$value,
          2*2 + 2*fit_loglogistic$value)
```

```r
)

model_comparison$Delta_AIC <- model_comparison$AIC - min(model_comparison$AIC)
model_comparison
```

```
         Model    LogLik      AIC Delta_AIC
1      Weibull -136.6462 277.2924 0.1451952
2    Lognormal -136.5736 277.1472 0.0000000
3 Log-Logistic -136.6422 277.2844 0.1372052
```

We applied the AIC to compare the three fitted models. The Lognormal distribution achieved the smallest AIC value (277.15), with the Weibull and Log-logistic models very close behind (ΔAIC < 0.15). Since all ΔAIC values are below 2, all three models provide comparably good fits, with a slight preference for the Lognormal model.

v) We will now check the hypothesis that the data follow an exponential distribution, which corresponds to a Weibull model with shape parameter $k = 1$

```r
# Inverse Hessian (approximate covariance matrix of (log k, log lambda))
cov_log <- solve(fit_weibull$hessian)

# Variance of log(k)
var_logk <- cov_log[1, 1]
se_logk  <- sqrt(var_logk)

# Delta method: k = exp(logk)
k_hat <- k_hat_weibull
se_k  <- k_hat * se_logk

# 95% CI for k
alpha <- 0.05
z <- qnorm(1 - alpha/2)

ci_k_lower <- k_hat - z * se_k
ci_k_upper <- k_hat + z * se_k

c(k_hat = k_hat, lower = ci_k_lower, upper = ci_k_upper)
```

```
   k_hat    lower    upper
2.576593 1.552022 3.601163
```

Using the observed information matrix from the Weibull ML fit, we constructed a 95% confidence interval for the shape parameter. The estimate was $\hat{k} = 2.5766$ with a 95% CI of $[1.5520, \; 3.6012]$. Since the value $k = 1$ does not lie within this interval, we reject the null hypothesis at the 5% significance level. Therefore, the exponential model is not adequate for these data, and a general Weibull model provides a significantly better fit.

(C)

In this segment we will analyze a dataset of 77 lung patients with 43 of them receiving improved lung therapy ($= 0$) and 34 of them did not ($= 1$).

i) We start with loading the data and computing KM estimates:

```
data <- read.table("/home/kontadak/Documents/File Hub/Semfe Grad/Survival Analysis/Series 2/1

# Check the data
head(data)
```

```
  id Group   t c
1  1     0  59 1
2  2     0 514 1
3  3     0 313 1
4  4     0 631 1
5  5     0 107 1
6  6     0  71 1
```

```
str(data)
```

```
'data.frame':   77 obs. of  4 variables:
 $ id   : int  1 2 3 4 5 6 7 8 9 10 ...
 $ Group: int  0 0 0 0 0 0 0 0 0 0 ...
 $ t    : int  59 514 313 631 107 71 583 91 66 95 ...
 $ c    : int  1 1 1 1 1 1 1 1 1 1 ...
```

```
# Summary by group
cat("Sample size by group:\n")
```

```
Sample size by group:
```

```r
print(table(data$Group))
```

```
 0  1 
43 34 
```

```r
cat("\nCensoring status by group:\n")
```

```
Censoring status by group:
```

```r
print(table(data$Group, data$c))
```

```
   
     0  1
  0  3 40
  1  2 32
```

```r
# Separate by group
group0 <- data[data$Group == 0, ]
group1 <- data[data$Group == 1, ]

# Manual Kaplan-Meier function
compute_km <- function(time, status) {
  n <- length(time)
  ord <- order(time)
  time_sorted <- time[ord]
  status_sorted <- status[ord]

  # Get unique event times
  event_times <- unique(time_sorted[status_sorted == 1])

  km_surv <- numeric(length(event_times))
  km_times <- event_times

  for (i in seq_along(event_times)) {
    t <- event_times[i]
    n_risk <- sum(time_sorted >= t)
    n_event <- sum(time_sorted == t & status_sorted == 1)
```

```r
    if (i == 1) {
      km_surv[i] <- 1 - n_event/n_risk
    } else {
      km_surv[i] <- km_surv[i-1] * (1 - n_event/n_risk)
    }
  }

  list(time = km_times, surv = km_surv)
}

# Compute KM for both groups
km0 <- compute_km(group0$t, group0$c)
km1 <- compute_km(group1$t, group1$c)

# Print tables
cat("Kaplan-Meier for Group 0 (Enhanced therapy):\n")
```

```
Kaplan-Meier for Group 0 (Enhanced therapy):
```

```r
km_table0 <- data.frame(Time = km0$time, Survival = km0$surv)
print(head(km_table0, 10))
```

```
   Time  Survival
1     5 0.9767442
2    13 0.9534884
3    16 0.9302326
4    26 0.9069767
5    32 0.8837209
6    33 0.8604651
7    48 0.8372093
8    56 0.8139535
9    59 0.7674419
10   62 0.7441860
```

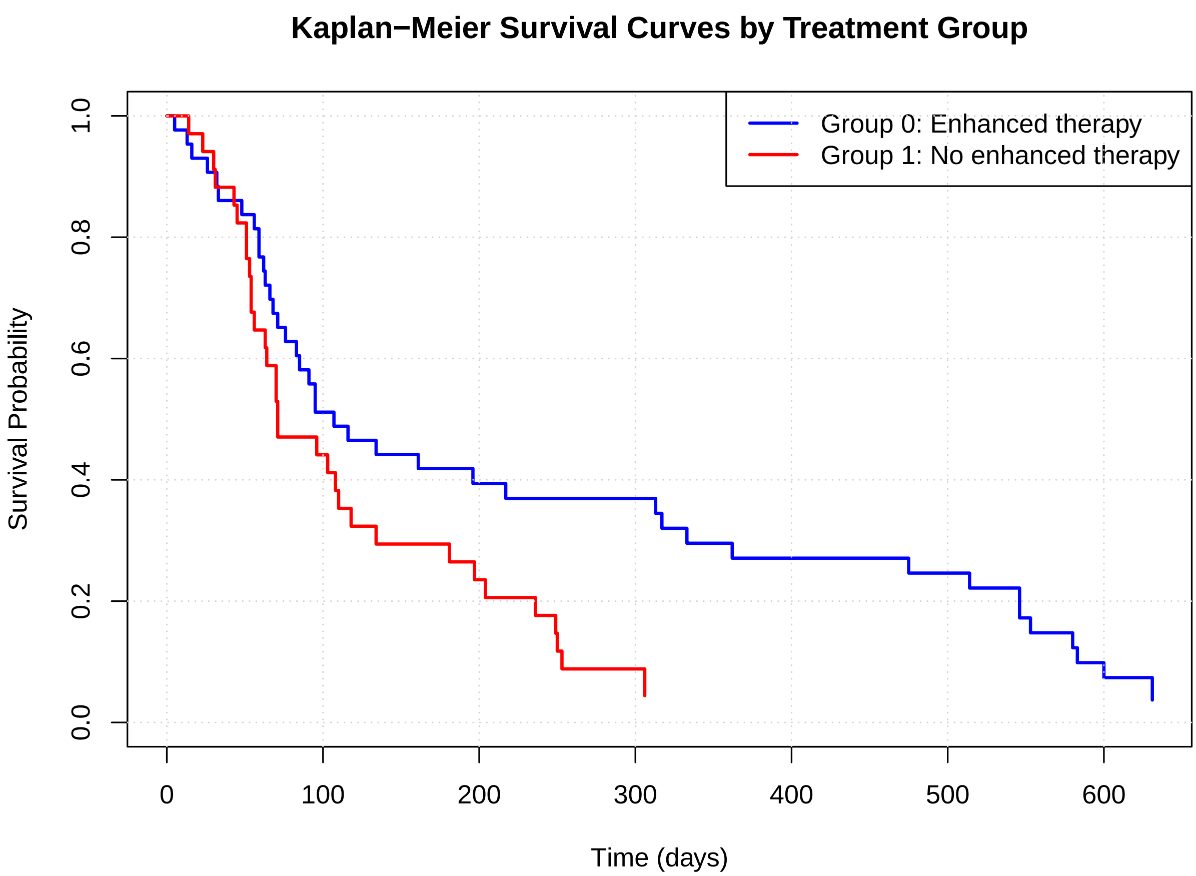
```r
cat("\nKaplan-Meier for Group 1 (No enhanced therapy):\n")
```

```

Kaplan-Meier for Group 1 (No enhanced therapy):
```

```r
km_table1 <- data.frame(Time = km1$time, Survival = km1$surv)
print(head(km_table1, 10))
```

```
   Time  Survival
1    14 0.9705882
2    23 0.9411765
3    30 0.9117647
4    31 0.8823529
5    43 0.8529412
6    45 0.8235294
7    51 0.7647059
8    53 0.7352941
9    54 0.6764706
10   56 0.6470588
```

```r
plot(c(0, km0$time), c(1, km0$surv),
     type = "s", col = "blue", lwd = 2,
     xlab = "Time (days)",
     ylab = "Survival Probability",
     main = "Kaplan-Meier Survival Curves by Treatment Group",
     ylim = c(0, 1))

lines(c(0, km1$time), c(1, km1$surv),
      type = "s", col = "red", lwd = 2)

legend("topright",
       legend = c("Group 0: Enhanced therapy", "Group 1: No enhanced therapy"),
       col = c("blue", "red"),
       lty = 1, lwd = 2)
grid()
```



ii) We will now run tests to compare if the two survival curves are significantly different.

```
# Log-rank test (manual implementation)
logrank_test <- function(time0, status0, time1, status1) {
  # Combine data
  all_times <- sort(unique(c(time0[status0 == 1], time1[status1 == 1])))

  O0 <- numeric(length(all_times))  # Observed events group 0
  O1 <- numeric(length(all_times))  # Observed events group 1
  E0 <- numeric(length(all_times))  # Expected events group 0
  E1 <- numeric(length(all_times))  # Expected events group 1

  for (i in seq_along(all_times)) {
    t <- all_times[i]

    # At risk
    n0 <- sum(time0 >= t)
    n1 <- sum(time1 >= t)
```

```r
  n_total <- n0 + n1

  # Events
  d0 <- sum(time0 == t & status0 == 1)
  d1 <- sum(time1 == t & status1 == 1)
  d_total <- d0 + d1

  O0[i] <- d0
  O1[i] <- d1
  E0[i] <- n0 * d_total / n_total
  E1[i] <- n1 * d_total / n_total
}

# Log-rank statistic
O0_total <- sum(O0)
E0_total <- sum(E0)

# Variance
V <- 0
for (i in seq_along(all_times)) {
  t <- all_times[i]
  n0 <- sum(time0 >= t)
  n1 <- sum(time1 >= t)
  n_total <- n0 + n1
  d_total <- O0[i] + O1[i]

  if (n_total > 1) {
    V <- V + (n0 * n1 * d_total * (n_total - d_total)) /
             (n_total^2 * (n_total - 1))
  }
}

# Test statistic
chi2 <- (O0_total - E0_total)^2 / V
p_value <- 1 - pchisq(chi2, df = 1)

list(
  statistic = chi2,
  p_value = p_value,
  observed_0 = O0_total,
  expected_0 = E0_total
)
```

```r
}

# Perform log-rank test
lr_result <- logrank_test(group0$t, group0$c, group1$t, group1$c)

cat("Log-Rank Test:\n")
```

```
Log-Rank Test:
```

```r
cat(sprintf("  Chi-square statistic: %.4f\n", lr_result$statistic))
```

```
  Chi-square statistic: 6.1999
```

```r
cat(sprintf("  p-value: %.4f\n", lr_result$p_value))
```

```
  p-value: 0.0128
```

```r
cat(sprintf("  Observed events (Group 0): %.0f\n", lr_result$observed_0))
```

```
  Observed events (Group 0): 40
```

```r
cat(sprintf("  Expected events (Group 0): %.2f\n", lr_result$expected_0))
```

```
  Expected events (Group 0): 49.21
```

```r
if (lr_result$p_value < 0.05) {
  cat("\n  Conclusion: Significant difference between groups (p < 0.05)\n")
} else {
  cat("\n  Conclusion: No significant difference between groups (p >= 0.05)\n")
}
```

```

  Conclusion: Significant difference between groups (p < 0.05)
```

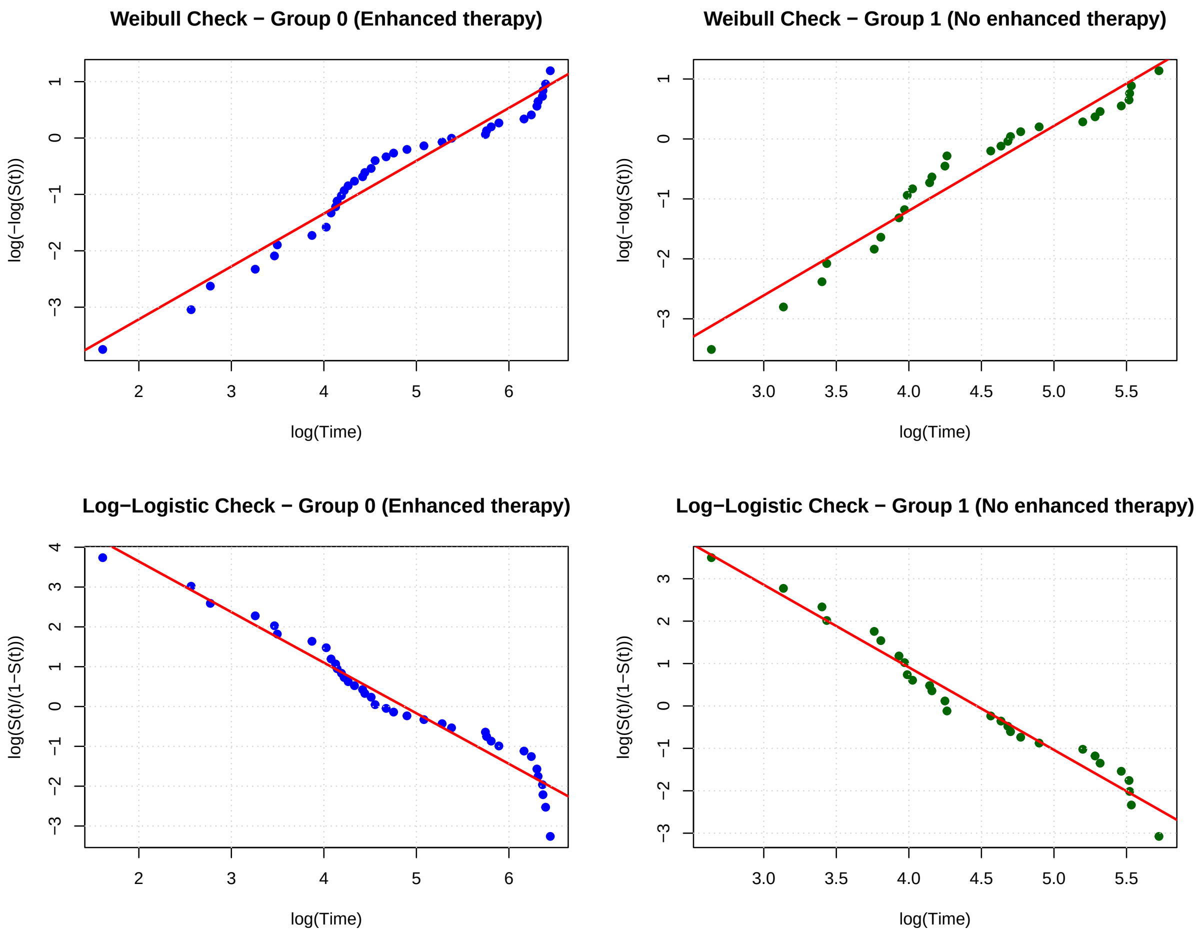
There is a significant difference between the two groups ($p = 0.0128 < 0.05$) Group 0 (enhanced therapy) had fewer deaths than expected (40 observed vs 49.21 expected), suggesting the enhanced therapy is helping patients survive longer.

iii)

```r
# Compute KM estimates
compute_km <- function(time, status) {
  n <- length(time)
  ord <- order(time)
  time_sorted <- time[ord]
  status_sorted <- status[ord]

  event_times <- unique(time_sorted[status_sorted == 1])
  km_surv <- numeric(length(event_times))

  for (i in seq_along(event_times)) {
    t <- event_times[i]
    n_risk <- sum(time_sorted >= t)
    n_event <- sum(time_sorted == t & status_sorted == 1)

    if (i == 1) {
      km_surv[i] <- 1 - n_event/n_risk
    } else {
      km_surv[i] <- km_surv[i-1] * (1 - n_event/n_risk)
    }
  }

  list(time = event_times, surv = km_surv)
}

# Separate by group
group0 <- data[data$Group == 0, ]
group1 <- data[data$Group == 1, ]

km0 <- compute_km(group0$t, group0$c)
km1 <- compute_km(group1$t, group1$c)

# Remove S(t) = 1 and S(t) = 0 for log transformations
filter_km <- function(km) {
  valid <- km$surv > 0 & km$surv < 1
  list(time = km$time[valid], surv = km$surv[valid])
}

km0_filtered <- filter_km(km0)
km1_filtered <- filter_km(km1)

# Create diagnostic plots
```

```r
par(mfrow = c(2, 2))

# WEIBULL CHECK: log(-log(S(t))) vs log(t) should be linear
# For Weibull: S(t) = exp(-(t/ )^k)
# log(-log(S(t))) = k*log(t) - k*log( )

# Group 0
plot(log(km0_filtered$time), log(-log(km0_filtered$surv)),
     pch = 19, col = "blue",
     xlab = "log(Time)", ylab = "log(-log(S(t)))",
     main = "Weibull Check - Group 0 (Enhanced therapy)")
abline(lm(log(-log(km0_filtered$surv)) ~ log(km0_filtered$time)), col = "red", lwd = 2)
grid()

# Group 1
plot(log(km1_filtered$time), log(-log(km1_filtered$surv)),
     pch = 19, col = "darkgreen",
     xlab = "log(Time)", ylab = "log(-log(S(t)))",
     main = "Weibull Check - Group 1 (No enhanced therapy)")
abline(lm(log(-log(km1_filtered$surv)) ~ log(km1_filtered$time)), col = "red", lwd = 2)
grid()

# LOG-LOGISTIC CHECK: log(S(t)/(1-S(t))) vs log(t) should be linear
# For Log-Logistic: S(t) = 1/(1 + (t/ )^ )
# log(S/(1-S)) = - *log(t) +  *log( )

# Group 0
odds0 <- km0_filtered$surv / (1 - km0_filtered$surv)
plot(log(km0_filtered$time), log(odds0),
     pch = 19, col = "blue",
     xlab = "log(Time)", ylab = "log(S(t)/(1-S(t)))",
     main = "Log-Logistic Check - Group 0 (Enhanced therapy)")
abline(lm(log(odds0) ~ log(km0_filtered$time)), col = "red", lwd = 2)
grid()

# Group 1
odds1 <- km1_filtered$surv / (1 - km1_filtered$surv)
plot(log(km1_filtered$time), log(odds1),
     pch = 19, col = "darkgreen",
     xlab = "log(Time)", ylab = "log(S(t)/(1-S(t)))",
     main = "Log-Logistic Check - Group 1 (No enhanced therapy)")
abline(lm(log(odds1) ~ log(km1_filtered$time)), col = "red", lwd = 2)
```

```r
grid()
```




```r
par(mfrow = c(1, 1))
```

Both distributions (Weibull and Log-Logistic) seem to fit the data well for both groups! We'll use AIC to decide which is better.

```r
# Log-Logistic negative log-likelihood
negloglik_loglogistic <- function(par, time, status) {
  alpha <- exp(par[1])
  beta  <- exp(par[2])

  t <- time
  d <- status
```

```r
  logf <- log(alpha) - log(beta) + (alpha - 1)*log(t/beta) -
          2*log(1 + (t/beta)^alpha)

  logS <- -log(1 + (t/beta)^alpha)

  -sum(d * logf + (1 - d) * logS)
}

# Fit Group 0
fit_ll_g0 <- optim(
  par = c(log(1), log(median(group0$t))),
  fn = negloglik_loglogistic,
  time = group0$t,
  status = group0$c,
  hessian = TRUE,
  method = "L-BFGS-B",
  lower = c(log(1e-6), log(1e-6)),
  upper = c(log(1e6),  log(1e6))
)

alpha_g0 <- exp(fit_ll_g0$par[1])
beta_g0 <- exp(fit_ll_g0$par[2])

# Median for Log-Logistic: median =
median_ll_g0 <- beta_g0

cat("Group 0 (Enhanced therapy) - Log-Logistic:\n")
```

```
Group 0 (Enhanced therapy) - Log-Logistic:
```

```r
cat(sprintf("    (shape) = %.4f\n", alpha_g0))
```

```
    (shape) = 1.3421
```

```r
cat(sprintf("    (scale) = %.4f\n", beta_g0))
```

```
    (scale) = 138.4817
```

```r
cat(sprintf("  Median survival = %.2f days\n", median_ll_g0))
```

```
  Median survival = 138.48 days
```

```r
cat(sprintf("  Log-likelihood = %.2f\n", -fit_ll_g0$value))
```

```
  Log-likelihood = -261.18
```

```r
cat(sprintf("  AIC = %.2f\n", 2*2 + 2*fit_ll_g0$value))
```

```
  AIC = 526.36
```

```r
# Fit Group 1
fit_ll_g1 <- optim(
  par = c(log(1), log(median(group1$t))),
  fn = negloglik_loglogistic,
  time = group1$t,
  status = group1$c,
  hessian = TRUE,
  method = "L-BFGS-B",
  lower = c(log(1e-6), log(1e-6)),
  upper = c(log(1e6),  log(1e6))
)

alpha_g1 <- exp(fit_ll_g1$par[1])
beta_g1 <- exp(fit_ll_g1$par[2])
median_ll_g1 <- beta_g1

cat("\nGroup 1 (No enhanced therapy) - Log-Logistic:\n")
```

```

Group 1 (No enhanced therapy) - Log-Logistic:
```

```r
cat(sprintf("    (shape) = %.4f\n", alpha_g1))
```

```
    (shape) = 2.0326
```

```r
cat(sprintf("    (scale) = %.4f\n", beta_g1))
```

```
    (scale) = 88.7508
```

```r
cat(sprintf("  Median survival = %.2f days\n", median_ll_g1))
```

```
  Median survival = 88.75 days
```

```r
cat(sprintf("  Log-likelihood = %.2f\n", -fit_ll_g1$value))
```

```
  Log-likelihood = -183.76
```

```r
cat(sprintf("  AIC = %.2f\n", 2*2 + 2*fit_ll_g1$value))
```

```
  AIC = 371.53
```

```r
# Weibull negative log-likelihood
negloglik_weibull <- function(par, time, status) {
  k      <- exp(par[1])
  lambda <- exp(par[2])

  t <- time
  d <- status

  logf <- d * ( log(k) - log(lambda) + (k - 1)*log(t/lambda) - (t/lambda)^k )
  logS <- (1 - d) * ( -(t/lambda)^k )

  out <- -(sum(logf) + sum(logS))
  if (!is.finite(out)) return(1e100)
  out
}

# Fit Weibull for Group 0
fit_weibull_g0 <- optim(
  par = c(log(1), log(median(group0$t))),
  fn = negloglik_weibull,
  time = group0$t,
  status = group0$c,
```

```r
  hessian = TRUE,
  method = "L-BFGS-B",
  lower = c(log(1e-6), log(1e-6)),
  upper = c(log(1e6),  log(1e6))
)

k_g0      <- exp(fit_weibull_g0$par[1])
lambda_g0 <- exp(fit_weibull_g0$par[2])
median_g0 <- lambda_g0 * (log(2))^(1/k_g0)

# Fit Weibull for Group 1
fit_weibull_g1 <- optim(
  par = c(log(1), log(median(group1$t))),
  fn = negloglik_weibull,
  time = group1$t,
  status = group1$c,
  hessian = TRUE,
  method = "L-BFGS-B",
  lower = c(log(1e-6), log(1e-6)),
  upper = c(log(1e6),  log(1e6))
)

k_g1      <- exp(fit_weibull_g1$par[1])
lambda_g1 <- exp(fit_weibull_g1$par[2])
median_g1 <- lambda_g1 * (log(2))^(1/k_g1)
```

```r
# Create comparison table
comparison <- data.frame(
  Group = c("Group 0", "Group 0", "Group 1", "Group 1"),
  Model = c("Weibull", "Log-Logistic", "Weibull", "Log-Logistic"),
  LogLik = c(-fit_weibull_g0$value, -fit_ll_g0$value,
             -fit_weibull_g1$value, -fit_ll_g1$value),
  AIC = c(2*2 + 2*fit_weibull_g0$value, 2*2 + 2*fit_ll_g0$value,
          2*2 + 2*fit_weibull_g1$value, 2*2 + 2*fit_ll_g1$value),
  Median = c(median_g0, median_ll_g0, median_g1, median_ll_g1)
)

comparison$Delta_AIC <- NA
comparison$Delta_AIC[1:2] <- comparison$AIC[1:2] - min(comparison$AIC[1:2])
comparison$Delta_AIC[3:4] <- comparison$AIC[3:4] - min(comparison$AIC[3:4])

print(comparison)
```

```
    Group        Model    LogLik      AIC    Median Delta_AIC
1 Group 0      Weibull -260.8308 525.6616 165.27098 0.0000000
2 Group 0 Log-Logistic -261.1803 526.3605 138.48167 0.6988699
3 Group 1      Weibull -185.1041 374.2082 102.06389 2.6783868
4 Group 1 Log-Logistic -183.7649 371.5298  88.75079 0.0000000
```

```r
cat("\n", strrep("=", 60), "\n")
```

```
 ============================================================
```

```r
cat("AIC INTERPRETATION:\n")
```

```
AIC INTERPRETATION:
```

```r
cat("Lower AIC = Better fit\n")
```

```
Lower AIC = Better fit
```

```r
cat("Delta_AIC = Difference from best model in each group\n\n")
```

```
Delta_AIC = Difference from best model in each group
```

```r
# Group 0 best model
best_g0 <- comparison$Model[which.min(comparison$AIC[1:2])]
cat(sprintf("Group 0: Best model is %s\n", best_g0))
```

```
Group 0: Best model is Weibull
```

```r
# Group 1 best model
best_g1 <- comparison$Model[2 + which.min(comparison$AIC[3:4])]
cat(sprintf("Group 1: Best model is %s\n", best_g1))
```

```
Group 1: Best model is Log-Logistic
```

For group 0, the Weibull model has the lowest AIC (525.66), outperforming the Log-Logistic model (AIC = 526.36). The Δ-AIC between the two models is 0.70, which is small, indicating that both models fit reasonably well. However, since the Weibull model achieves the minimum AIC, it is the preferred model for Group 0.

Group 1 (No enhanced therapy) The Log-Logistic model has the lowest AIC (371.53), clearly outperforming the Weibull model (AIC = 374.21). The Δ-AIC is 2.68, which provides moderate evidence in favor of the Log-Logistic model. Thus, the Log-Logistic distribution is the preferred model for Group 1.

iv) We are now going

```
# Weibull median survival for each group
cat("Weibull MLE Results:\n\n")
```

```
Weibull MLE Results:
```

```
cat("Group 0 (Enhanced therapy):\n")
```

```
Group 0 (Enhanced therapy):
```

```
cat(sprintf("  Shape (k)    = %.4f\n", k_g0))
```

```
  Shape (k)    = 0.9339
```

```
cat(sprintf("  Scale ( )    = %.4f\n", lambda_g0))
```

```
  Scale ( )    = 244.7065
```

```
cat(sprintf("  Median (days) = %.2f\n\n", median_g0))
```

```
  Median (days) = 165.27
```

```
cat("Group 1 (No enhanced therapy):\n")
```

```
Group 1 (No enhanced therapy):
```

```r
cat(sprintf("  Shape (k)    = %.4f\n", k_g1))
```

```
  Shape (k)    = 1.3176
```

```r
cat(sprintf("  Scale ( )    = %.4f\n", lambda_g1))
```

```
  Scale ( )    = 134.7972
```

```r
cat(sprintf("  Median (days) = %.2f\n\n", median_g1))
```

```
  Median (days) = 102.06
```

```r
# Comparison
if (median_g0 > median_g1) {
  cat("Conclusion: Group 0 remains on oxygen longer on average.\n")
} else {
  cat("Conclusion: Group 1 remains on oxygen longer on average.\n")
}
```

```
Conclusion: Group 0 remains on oxygen longer on average.
```

Under the assumption that the Weibull model adequately describes the survival distribution for both groups, we estimated the shape and scale parameters separately for the enhanced-therapy group (Group 0) and the non-enhanced group (Group 1). Using these estimates, we computed the median survival time for each group.

Group 0 exhibited a median oxygen-support duration of $\tilde{t}_0 = 165.27$ days, whereas Group 1 had a median duration of $\tilde{t}_1 = 102.06$ days.

Since the median survival under the Weibull model is larger for Group 0, we conclude that patients receiving enhanced lung therapy tend to remain on oxygen for a longer period before disconnection. This finding is consistent with the earlier log-rank test, which also indicated improved survival for the enhanced-therapy group.

Patients with enhanced therapy survive ~60 days longer on average.